MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	CSM-A11

.

Title of the Project: Career-Oriented Resource Planning and Personalized Recommendation Using Deep Reinforcement Learning

Abstract

Career-oriented resource planning and personalized recommendation have become essential in today's competitive educational and professional environments, helping individuals make informed career decisions and develop the skills required to achieve their goals. Traditional career guidance systems often face challenges such as limited personalization, inadequate skill gap analysis, inefficient learning resource allocation, and difficulties in adapting recommendations to users' changing interests and career objectives. These limitations can affect career planning, skill development, and professional growth. This project presents the design and development of a **Career-Oriented Resource Planning and Personalized Recommendation System Using Deep Reinforcement Learning** that provides intelligent, adaptive, and personalized career guidance.

The proposed system analyzes users' educational qualifications, technical skills, interests, career preferences, and professional goals to recommend suitable career paths, learning resources, skill development courses, and relevant job opportunities. To enhance the recommendation process, the system integrates **Deep Reinforcement Learning (DRL)** techniques that enable the model to learn from user interactions, feedback, and learning progress. A reward-based mechanism is used to improve the relevance of recommendations and adapt them according to individual needs and long-term career objectives.

The platform also incorporates resource planning and skill gap analysis to identify the competencies required for selected career paths. Based on these insights, it suggests appropriate learning materials, prioritizes skill development activities, and generates personalized learning plans according to users' available time and learning requirements. The system can update recommendations dynamically as users acquire new skills or modify their preferences. User profiles, progress tracking, and recommendation analytics further support effective career management and continuous improvement.

By combining personalized recommendation techniques, intelligent resource planning, and deep reinforcement learning, the proposed system aims to provide a scalable, adaptive, and user-friendly career guidance platform. It can be applied in educational institutions, career counseling centers, and professional development environments, helping students and job seekers identify suitable career opportunities, utilize learning resources efficiently, bridge skill gaps, and make better-informed decisions for their future careers.

MAJOR PROJECT REGISTRATION FORM

Date:

References

1. ***Li, D., Peng, L., & Zhang, J. (2026).*** *Design and Validation of Personalized Career Path Planning Algorithm Based on Deep Reinforcement Learning (DRL). Procedia Computer Science, 281, 236–244.*
*DOI: **10.1016/j.procs.2026.04.212***

2. ***Lian, Y., & He, F. (2026).*** *A Reinforcement Learning-Based Framework for Personalized Career Recommendations Among College Students. Journal of Advanced Computational Intelligence and Intelligent Informatics, 30(5), 1500–1514.*
DOI: 10.20965/jaciii.2026.p1500

3. ***Chen, X., Yao, L., McAuley, J., Zhou, G., & Wang, X. (2023).*** *Deep Reinforcement Learning in Recommender Systems: A Survey and New Perspectives. Knowledge-Based Systems, 264, Article 110335.*
DOI: 10.1016/j.knosys.2023.110335

4. ***Kokkodis, M., & Ipeirotis, P. G. (2021).*** *Demand-Aware Career Path Recommendations: A Reinforcement Learning Approach. Management Science, 67(7), 4362–4383.*
DOI: 10.1287/mnsc.2020.3727.

5. ***Nabizadeh, A. H., Leal, J. P., Rafsanjani, H. N., & Shah, R. R. (2020).*** *Learning Path Personalization and Recommendation Methods: A Survey of the State-of-the-Art. Expert Systems with Applications, 159, Article 113596.*
DOI: 10.1016/j.eswa.2020.113596

6. ***Lin, Y., Liu, Y., Lin, F., Zou, L., Wu, P., Zeng, W., Chen, H., & Miao, C. (2024).*** *A Survey on Reinforcement Learning for Recommender Systems. IEEE Transactions on Neural Networks and Learning Systems, 35(10), 13164–13184.*
DOI: 10.1109/TNNLS.2023.3280161

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A4229	Cherukuri Supriya Sundari	
2	23BQ1A4239	Dhanireddy Vinay Kumar Reddy	
3	23BQ1A4222	Bodepudi Naga Soumya	
4	23BQ1A4248	Galla Shanmuka Venkata Sai Santosh	

Project Guide

Name: Mrs. M. Lavanya Signature:

MAJOR PROJECT REGISTRATION FORM

Date: